



Deep learning in engineering technology

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Chapter 05

Neural network review



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Exercise



Exercise #1

Define what machine learning is and how it is different from classical statistics. Provide the fundamental machine learning tasks, briefly define each and give an example of where each of them can be used.

Hints: Compare the following items.

1) Main Goal
2) Focus
3) Data Size
4) Model Complexity
5) Success Measure
6) Example





Exercise #2

a) Compare **Supervised Learning** and **Unsupervised Learning** in terms of their data requirements and core objectives.
b) Identify and define the main sub-categories/algorithms under each paradigm:
    - Classification
    - Clustering
    - Regression
    - Association
    - Dimension Reduction
c) Provide a simple, real-world everyday analogy for each sub-category to illustrate how it works.





Exercise #3

Identify whether each scenario is an example of supervised learning, unsupervised learning, or reinforcement learning. Explain your answer.

a) A neural network learns to recognize handwritten digits using a dataset of images where the correct digit label is provided for each image.

b) A robot learns to play a video game by trying different actions and improving its score over time. It does not know the game's rules beforehand.

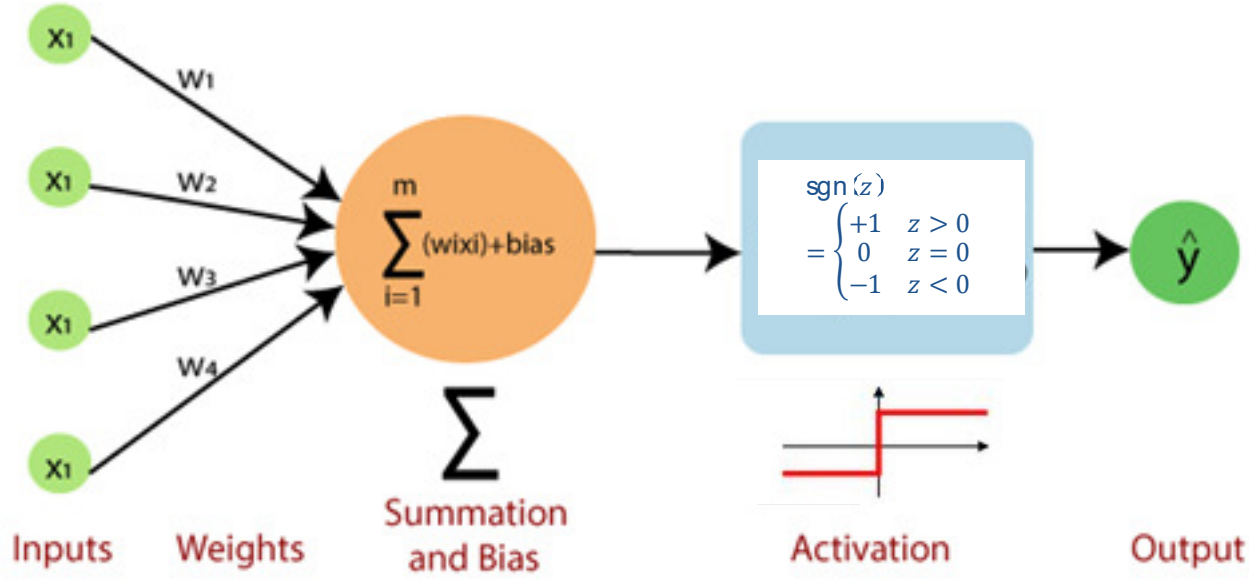


Exercise #4

We have four data points in the dataset:

1) ($\boldsymbol{x}1 = [1, 1, 2, 3]$, $y1 = 1$)
2) ($\boldsymbol{x}2 = [1, 1, 1, 1]$, $y2 = 1$)
3) ($\boldsymbol{x}3 = [1, -1, -1, -1]$, $y3 = -1$)
4) ($\boldsymbol{x}4 = [1, -2, -2, -2]$, $y4 = -1$).

Consider a Perceptron without bias and use signum function for4 activation.



$$\hat{y} = \text{sgn}(w^T x_i)$$

with update rule for a mini-batch:

$$w_{\text{new}} = w_{\text{old}} + \eta \sum_i (y_i - \hat{y}_i) x_i$$

$$\Delta w = \eta \sum_{i \in B} (y_i - \hat{y}_i)\, \mathbf{x}_i$$

Do two iterations of the mini-batch training of Perceptron with batches:

- mini-batch for iteration 1: data points 1 and 4
- mini-batch for iteration 2: data points 2 and 3

Use the learning rate $\eta = 0.01$. Let the initial weights be $w(0) = [0, 0.1, 0.2, -0.3]$.



Exercise #5

Modify exercise #4 as follows:

1) Loss function is changed to mean square error.
2) The activation function is the **Identity Function** (Linear Activation):

$$f(z) = z$$

Where $z = \mathbf{w}^T\mathbf{x} = w_0 + w_1x_1 + w_2x_2 + w_3x_3$.

During training, the raw linear combination $\hat{y}_i = \mathbf{w}^T\mathbf{x}_i$ is passed directly to the loss function without thresholding or discretization.

3) Keep the batch update as:

$$\Delta\mathbf{w} = \eta \sum_{i \in B} (y_i - \mathbf{w}^T\mathbf{x}_i)\,\mathbf{x}_i = \sum_{i \in B} (y_i - \hat{y}_i)\,\mathbf{x}_i$$





Exercise #6

Consider a neural network with a single real-valued input $x \in \mathrm{R}$ and a single output $y \in \mathrm{R}$. The network architecture is defined as follows:

$$y = v_1 \cdot \tanh(w_1 x) + v_2 \cdot \tanh(w_2 x)$$

where $\mathrm{w} = [w_1, w_2] \in R^2$ are the weights of the first layer, $\mathrm{v} = [v_1, v_2] \in R^2$ are the weights of the output layer, and $\varphi(z) = \tanh(z)$ is the activation function used in the hidden layer. Given a training sample $(x^{(1)}, y_{\text{true}})$, we use the mean squared error (MSE) loss function:

$$L = \frac{1}{2}(y - y_{true})^2$$

Compute the following quantities:

a) Derive expressions for the gradients $\frac{\partial L}{\partial v_1}$ and $\frac{\partial L}{\partial v_2}$.

b) Derive expressions for the gradients $\frac{\partial L}{\partial w_1}$ and $\frac{\partial L}{\partial w_2}$.

c) Given $x^{(1)} = 1$, $y_{\text{true}} = 0$, $w_1 = w_2 = 1$, and $v_1 = v_2 = 1$, compute the numerical values of all the gradients obtained in the backward pass.

Hints: $\frac{d}{dz}\tanh(z) = 1 - \tanh^2(z)$



Exercise #7

1. Which issue does subword tokenization primarily address compared to traditional word-level tokenization?
   a) Quadratic attention complensity
   b) Catastrophic forgetting
   c) Vanishing gradient problem
   d) Out of vocabulary (OOV) words in the sample or library.





Exercise #7

2. In the Byte-Pair Encoding (BPE) algorithm, what criterion determines which pair of adjacent tokens is merged in each iteration?
   a) Lowest entropy loss which indicate accuracy of the prediction of a model.
   b) Highest co-occurrence frequency
   c) Maximized language model likelihood
   d) Minimization of Unigram loss which is a loss function or regularization term that encourages a model's predicted token distribution to align with a predefined unigram distribution.





Exercise #7

3. How does the Unigram language model tokenization algorithm differ fundamentally from Byte-Pair Encoding (BPE) during vocabulary construction?
   a) BPE uses characters; Unigram uses full words.
   b) BPE merges units while Unigram prunes units
   c) BPE requires dynamic programming while Unigram does not
   d) BPE is probabilistic while Unigram is deterministic.





Exercise #8

Compare the following features of RNN and LSTM unit. Hence, draw a chart diagram to illustrate the difference between RNN and LSTM unit.

1) Handles long-term dependencies
2) Computational complexity.
3) Memory efficiency
4) Speed of training
5) Number of gates.
6) Use cases

